

Bangladesh Weather Advisor Budget Proposal

GENIE.AI — GenAI for Good Challenge

Climate CHP Team | May 2026

6-Month Pilot: \$10,000 | 5-Year Roadmap: \$952,000

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1. Executive Summary

The Bangladesh Weather Advisor is a GenAI-powered agricultural advisory system that delivers localized, actionable weather and crop guidance to smallholder farmers via SMS, IVR, and web interfaces. This budget proposal outlines the financial requirements for a 6-month pilot phase (\$10,000) and a 5-year national scaling roadmap (\$952,000 cumulative).

The system leverages open satellite data (CHIRPS, ERA5-Land, MODIS), government sources (BMD, BAMIS), and the GENIE.AI Retrieval-Augmented Generation platform to generate district-specific advisories in Bangla for 64 districts across Bangladesh.

Key Financial Highlights

- ▶ 6-Month Pilot: \$10,000 total — targeting 500 farmers in 2 districts
- ▶ Year 1 Full Scale: \$50,000 — expansion to 2 divisions (14 districts)
- ▶ Year 5 National: \$362,000 — serving 200,000+ farmers across all 8 divisions
- ▶ Cost per Farmer at Scale: ~\$1.81/farmer/year (Year 5)
- ▶ Primary Cost Drivers: SMS delivery (35-40%) and personnel (25-30%)

2. 6-Month Pilot Budget

Total Allocation: \$10,000

The pilot phase targets 2 districts (Rajshahi and Rangpur divisions) with 500 registered farmers over 6 months, covering one complete Aman rice growing season.

2.1 Pilot Budget Summary

Category	Amount	% of Total	Type
Personnel (Part-time)	\$2,800	28%	Fixed
SMS Gateway & Telecom	\$2,500	25%	Variable
Cloud Infrastructure	\$1,800	18%	Fixed
Weather Data Subscriptions	\$1,200	12%	Fixed
Testing, QA & Contingency	\$1,700	17%	Mixed
TOTAL	\$10,000	100%	

2.2 Detailed Pilot Line Items

Personnel — \$2,800

Role	Hours/Month	Rate	Months	Subtotal
Project Lead (part-time)	20	\$15/hr	6	\$1,800
Data Engineer (part-time)	10	\$12/hr	6	\$720
Local Coordinator (Bangladesh)	—	Stipend	6	\$280
			Total	\$2,800

SMS Gateway & Telecom — \$2,500

Item	Unit Cost	Quantity	Subtotal
SMS Gateway Setup (Twilio/local)	—	1	\$200
Outbound SMS (advisories)	\$0.02/SMS	75,000	\$1,500
Inbound SMS (farmer queries)	\$0.01/SMS	10,000	\$100
Bangla Unicode SMS premium	\$0.005/SMS	75,000	\$375
SIM cards & local testing	—	—	\$325
		Total	\$2,500

Cloud Infrastructure — \$1,800

Service	Monthly Cost	Months	Subtotal
Google Cloud (Compute + Storage)	\$120	6	\$720
BigQuery (data warehouse)	\$50	6	\$300
GENIE.AI Platform Access	\$80	6	\$480

Domain & SSL + CDN	\$50	1	\$50
Monitoring (Uptime, Logs)	\$~42	6	\$250
		Total	\$1,800

Weather Data Subscriptions — \$1,200

Data Source	Cost Type	Subtotal
CHIRPS Rainfall (open)	Free	\$0
ERA5-Land Temperature (open)	Free	\$0
MODIS NDVI (open)	Free	\$0
BMD Premium API Access	Subscription	\$500
NOAA Climate Drivers	Free	\$0
OpenWeatherMap (backup forecasts)	API Plan	\$400
Data validation & QA tools	License	\$300
	Total	\$1,200

Testing, QA & Contingency — \$1,700

Item	Subtotal
User Acceptance Testing (UAT)	\$300
Field testing & farmer feedback sessions	\$400
Bug fixes & hotfix deployments	\$200
Translation & localization QA (Bangla)	\$200
Contingency reserve (unforeseen costs)	\$600
Total	\$1,700

3. 5-Year Scaling Roadmap

The following table summarizes the annual budget over a 5-year national scaling trajectory, from pilot completion through full national coverage serving 200,000+ farmers.

Year	Annual Budget	Cumulative	Key Milestones	Target Farmers
Pilot	\$10,000	\$10,000	2 districts, system validation	500
Year 1	\$50,000	\$60,000	Expand to 2 divisions	5,000
Year 2	\$100,000	\$160,000	IVR channels, ML models	20,000
Year 3	\$178,000	\$338,000	National coverage (all 64 districts)	80,000
Year 4	\$262,000	\$600,000	Advanced AI, crop insurance integration	140,000
Year 5	\$362,000	\$962,000	Full scale, 200K farmers, sustainability	200,000
Total	\$962,000			200,000+

Note: Cumulative total includes the \$10,000 pilot phase. Year-over-year growth reflects increasing SMS volumes, personnel, and infrastructure to support national coverage.

4. Year-by-Year Detailed Breakdown

4.1 Year 1 — \$50,000 (Expansion to 2 Divisions)

Focus: Expand from pilot to 14 districts across Rajshahi and Rangpur divisions. Hire dedicated part-time team. Establish partnerships with DAE extension offices.

Line Item	Fixed	Variable	Total	Notes
Personnel (2 part-time engineers)	\$12,000	—	\$12,000	20 hrs/wk each
Local field coordinators (2)	\$4,000	—	\$4,000	Stipend-based
SMS delivery (5,000 farmers)	—	\$9,000	\$9,000	~3 SMS/farmer/week
Cloud infrastructure	\$6,000	\$1,500	\$7,500	Scaled compute
GENIE.AI platform	\$4,800	—	\$4,800	Annual license
Weather data & APIs	\$3,000	—	\$3,000	BMD + backup sources
Training & capacity building	\$2,500	—	\$2,500	Farmer training sessions
Monitoring & evaluation	\$2,200	—	\$2,200	Impact assessment
Contingency (10%)	—	—	\$5,000	Risk buffer
YEAR 1 TOTAL	\$34,500	\$10,500	\$50,000	

4.2 Year 2 — \$100,000 (IVR Channels & ML Models)

Focus: Launch IVR (Interactive Voice Response) for non-literate farmers. Deploy ML-based yield prediction models. Expand to 4 divisions (32 districts).

Line Item	Fixed	Variable	Total	Notes
Personnel (3 engineers + PM)	\$28,000	—	\$28,000	Part-time to full-time
Local team (4 coordinators)	\$8,000	—	\$8,000	Division-level
SMS delivery (20,000 farmers)	—	\$16,000	\$16,000	Scaled volume
IVR system setup & operations	\$5,000	\$4,000	\$9,000	Bangla voice system
Cloud infrastructure	\$9,000	\$3,000	\$12,000	GPU for ML training
GENIE.AI platform	\$6,000	—	\$6,000	Enhanced tier
Weather data & APIs	\$4,000	—	\$4,000	Additional sources
ML model development	\$5,000	—	\$5,000	Yield prediction R&D
Training & outreach	\$3,000	—	\$3,000	Expanded program
Contingency (9%)	—	—	\$9,000	Risk buffer
YEAR 2 TOTAL	\$68,000	\$23,000	\$100,000	

4.3 Year 3 — \$178,000 (National Coverage)

Focus: Achieve national coverage across all 8 divisions (64 districts). Deploy advanced drought early warning system. Partner with government agricultural extension services.

Line Item	Fixed	Variable	Total	Notes
Personnel (5 team members)	\$48,000	—	\$48,000	Full-time core team

Local team (8 coordinators)	\$16,000	—	\$16,000	All 8 divisions
SMS delivery (80,000 farmers)	—	\$32,000	\$32,000	National scale
IVR operations	\$6,000	\$8,000	\$14,000	Expanded voice
Cloud infrastructure	\$14,000	\$6,000	\$20,000	Production-grade
GENIE.AI platform	\$8,000	—	\$8,000	Enterprise tier
Weather data & APIs	\$6,000	—	\$6,000	Premium sources
ML & AI development	\$8,000	—	\$8,000	Advanced models
Partnerships & integration	\$5,000	—	\$5,000	Govt. systems
M&E and impact studies	\$5,000	—	\$5,000	Academic research
Contingency (9%)	—	—	\$16,000	Risk buffer
YEAR 3 TOTAL	\$116,000	\$46,000	\$178,000	

4.4 Year 4 — \$262,000 (Advanced AI & Insurance Integration)

Focus: Integrate crop insurance recommendations. Deploy advanced AI for pest/disease prediction. Establish data marketplace for agricultural insights.

Line Item	Fixed	Variable	Total	Notes
Personnel (7 team members)	\$68,000	—	\$68,000	Expanded team
Local team (10 coordinators)	\$20,000	—	\$20,000	District-level leads
SMS delivery (140,000 farmers)	—	\$48,000	\$48,000	High volume
IVR & voice AI	\$10,000	\$12,000	\$22,000	AI-powered voice
Cloud infrastructure	\$20,000	\$10,000	\$30,000	Enterprise grade
GENIE.AI platform	\$12,000	—	\$12,000	Custom deployment
Weather & agri data	\$8,000	—	\$8,000	Premium + proprietary
AI/ML R&D	\$14,000	—	\$14,000	Pest/disease models
Insurance integration	\$8,000	—	\$8,000	API development
Partnerships & sustainability	\$8,000	—	\$8,000	Revenue model R&D
Contingency (10%)	—	—	\$24,000	Risk buffer
YEAR 4 TOTAL	\$168,000	\$70,000	\$262,000	

4.5 Year 5 — \$362,000 (Full Scale — 200K Farmers)

Focus: Reach 200,000 farmers nationally. Achieve financial sustainability through government partnerships, premium services, and data licensing. Full automation of advisory pipeline.

Line Item	Fixed	Variable	Total	Notes
Personnel (10 team members)	\$95,000	—	\$95,000	Full org structure
Local team (16 coordinators)	\$32,000	—	\$32,000	All divisions + HQ
SMS delivery (200,000 farmers)	—	\$68,000	\$68,000	Negotiated bulk rates
IVR & voice AI	\$12,000	\$18,000	\$30,000	Full voice platform
Cloud infrastructure	\$28,000	\$14,000	\$42,000	Multi-region HA
GENIE.AI platform	\$15,000	—	\$15,000	Enterprise + custom

Weather & agri data	\$10,000	—	\$10,000	Comprehensive sources
AI/ML operations	\$12,000	—	\$12,000	Model maintenance
Sustainability & revenue ops	\$10,000	—	\$10,000	Business development
Marketing & farmer acquisition	\$8,000	—	\$8,000	Growth campaigns
M&E and impact reporting	\$6,000	—	\$6,000	Annual reports
Contingency (9%)	—	—	\$34,000	Risk buffer
YEAR 5 TOTAL	\$238,000	\$100,000	\$362,000	

5. Cost Category Analysis

This section provides a cross-year view of major cost categories, illustrating how spending shifts from infrastructure-heavy (pilot) to operations-heavy (at scale).

Category	Pilot	Year 1	Year 2	Year 3	Year 4	Year 5
Personnel	\$2,800	\$16,000	\$36,000	\$64,000	\$88,000	\$127,000
SMS & Telecom	\$2,500	\$9,000	\$16,000	\$32,000	\$48,000	\$68,000
IVR / Voice	—	—	\$9,000	\$14,000	\$22,000	\$30,000
Cloud Infra	\$1,800	\$7,500	\$12,000	\$20,000	\$30,000	\$42,000
AI/ML Platform	—	\$4,800	\$11,000	\$16,000	\$26,000	\$27,000
Data Sources	\$1,200	\$3,000	\$4,000	\$6,000	\$8,000	\$10,000
Field Ops & Training	—	\$4,700	\$3,000	\$10,000	\$16,000	\$24,000
Contingency	\$1,700	\$5,000	\$9,000	\$16,000	\$24,000	\$34,000
TOTAL	\$10,000	\$50,000	\$100,000	\$178,000	\$262,000	\$362,000

6. Budget Justification Notes

Personnel Costs

Part-time roles during pilot shift to full-time as farmer base grows. Rates are competitive for Bangladesh tech market (\$8-15/hr for engineers). Local coordinators at stipend-based rates ensure ground-level farmer engagement.

SMS as Primary Channel

SMS remains the most reliable channel for rural Bangladesh where smartphone penetration is ~40% but basic phone coverage exceeds 95%. Cost of \$0.02/SMS for outbound Bangla Unicode messages reflects negotiated rates with local telecom providers (Grameenphone, Robi).

Cloud Infrastructure Scaling

Pilot uses Google Cloud free tier credits where possible. Costs scale with data processing volume (daily satellite processing for 64 districts) and user query loads. BigQuery provides cost-effective analytical storage at \$5/TB/month.

Weather Data Strategy

Primary data sources (CHIRPS, ERA5, MODIS) are open/free. Budget covers BMD premium API for real-time station data, backup forecast APIs, and data validation tools. No single-vendor dependency.

IVR Investment (Year 2+)

Interactive Voice Response serves non-literate farmers (~35% of target population). Initial setup cost of \$5,000 covers Bangla text-to-speech development. Per-call costs of \$0.03-0.05 make this cost-effective at scale.

Contingency Planning

Contingency ranges from 9-17% across years, higher during pilot (uncertainty) and Year 4 (insurance integration complexity). Unspent contingency rolls forward.

7. Risk & Contingency Planning

Risk	Likelihood	Impact	Mitigation	Budget Impact
Telecom price increase	Medium	High	Multi-provider contracts	+\$5,000/yr
BMD API unavailability	Medium	Medium	Backup data sources (OWM, GFS)	+\$2,000/yr
Low farmer adoption	Low	High	Field coordinators, radio campaigns	+\$3,000/yr
Cloud cost overrun	Low	Medium	Auto-scaling limits, reserved instances	+\$2,000/yr
Regulatory changes	Low	Medium	Legal counsel, govt. partnerships	+\$1,000/yr
Currency fluctuation (BDT/USD)	Medium	Low	Quarterly budget reviews	Absorbed

The contingency allocation (9-17% per year) is designed to absorb these risks without requiring additional funding requests. Quarterly budget reviews will reallocate unused contingency to high-priority growth areas.